

Morphometrical Analysis of the Short-Term Effects of Hypophysectomy and Food Restriction on Skeletal Muscle Fibers in Relation to Growth and Aging Changes in the Rat

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ABSTRACT: Muscle fiber growth is impaired by lack of hormonal influence. In the case of the gastrocnemius muscle of the male Wistar rat, ablation of the pituitary at age 60 days results in cessation of the growth of type 1 muscle fibers, and a reduction in the size of type 2 muscle fibers at 120 days of age, 60 days after hypophysectomy. The effects of food restriction, which are known to reduce pituitary hormone secretion, are not as great as hypophysectomy in inhibiting muscle fiber growth. Food restriction slowed the growth of type 1 muscle fibers (hypophysectomy abolished growth) and caused cessation of growth of type 2 muscle fibers (hypophysectomy caused atrophy).

KEY WORDS: Morphometry, Hypophysectomy, Food Restriction, Muscle Fiber Size, Type 1 Fibers, Type 2 Fibers, Gastrocnemius Muscle.

INTRODUCTION

The results of morphometrical analysis of gastrocnemius muscle fiber aging were recently reported by Shorey *et al.* (1988). Quantifying the changes that occur in muscle fiber size during the aging process in old control rats aged around 1000 days (33 months) enabled us to determine the extent of type 2 (fast) fiber atrophy and accompanying compensatory hypertrophy in the type 1 (posture) fibers.

Type 2 fiber atrophy is consistent with the hind-limb paralysis observed in many of the old control rats (Everitt *et al.*, 1985).

We have also examined the effects of long-term hypophysectomy and long-term food restriction on the aging process of skeletal muscle (Shorey *et al.*, 1988). It was concluded that for type 2 fibers there is a significant reduction in the mean fiber size which was indicative of atrophy between medium age (approx. 450 days) when whole body growth has ceased, and old age (approx. 1000 days) in the control rats. It was apparent that long-term hypophysectomy, and to a lesser extent long-term food restriction, reduced the normal process of atrophy of type 2 fibers and compensatory hypertrophy of the type 1 fibers. This was evident from reductions in the coefficient of variation, and the range of fiber size. The long-term hypophysectomized rats had smaller type 1 posture fibers in the gastrocnemius muscle than those in the control rats of the same age due to lack of growth.

Long-term hypophysectomy and food restriction over a period of 800-900 days (27-30 months) have been shown to retard gastrocnemius muscle fiber aging in the rat. The present study seeks to determine if these anti-aging effects recorded in old age are due to early changes occurring in the first two months after hypophysectomy and the introduction of food restriction, or whether they are due to the life long effects of hormone deficiency and reduced nutrient intake.

MATERIALS AND METHODS

Morphometrical analyses were carried out on 5 groups of rats namely 60 days-old controls, 90 days-old controls, 120 days-old controls, 120 days-old hypophysectomized (hypox) rats and 120 days-old food restricted (FR) rats. Each group comprised 4 rats from which the right and left gastrocnemius muscles were analysed. All rats were conventional non-inbred males of the University of Sydney Wistar strain; they were housed individually in wire bottom cages in a temperature controlled room at 25°C, with 12 hours of artificial light per day. Rats were fed pelleted food (Doust and Rabbidge, Sydney) of the following composition: protein 21.9%, fat 3.3%, fiber 3.6%, calcium 1.32%, phosphorus 0.88%, NaCl 0.5%. Food was available *ad libitum* except for the food restricted rats. Water was available *ad libitum* to all groups. Hypophysectomy was carried out when the rats were aged about 60 days, with a corresponding weight of 250-280 grams, using the technique of Koyama (1962), as modified by Everitt and Wyndham (1982). In order to control the effect of food, the food intake of food restricted intact rats was similar to that of the hypophysectomized rats (9g/day), about 50% of intact control intake. Food restriction was commenced when the rats were aged approximately 50 days and for 3 weeks they were fed decreasing amounts of food to bring about an accommodation of 9g per day at 70 days of age. This procedure is used to obviate the mortality caused by abrupt reduction of food intake at age 60 days in intact rats.

Rats were euthanized with anesthetic ether and the medial head of the gastrocnemius muscle was dissected from both legs, placed on a sheet of dental wax and any adhering fascia from the surrounding muscles was trimmed away. Each muscle was coated in OCT embedding compound for frozen tissue (Tissue Tek 4583, Miles Scientific, USA), quenched in super-cooled isopentane for 16s and subsequently stored in liquid nitrogen. Prior to section cutting the blocks of tissue were removed from liquid nitrogen storage, placed in a cryostat chamber and maintained at -25°C.

From each gastrocnemius muscle 9 serial sections were cut at a thickness of 10µm (level 1) followed by an advance of 1mm and the cutting of a similar number of sections (level 2). Three levels were cut in the mid-belly region of each muscle to enable the analysis of sections within levels and between levels within muscles. Measurements and analyses of skeletal muscle fibers were carried out as described by Shorey *et al.* (1988). Muscle sections were placed on chrome-alum gelatin-coated slides at room temperature and stored at -25°C before staining to demonstrate myofibrillar adenosine triphosphatase (ATPase)

activity at pH 9.4-10.0 (Padykula and Herman, 1955; Guth and Samaha, 1969). Type 1 fibers are lightly stained in section while type 2 fibers are darkly stained. Three random areas from each of three sections were photographed from 3 levels within each muscle. Fiber cross-sections were measured for area and equivalent circle diameter (ECD) which is computed as $2\sqrt{(\text{area}/\pi)}$ by the use of a fully automated Tracor Northern (TN 8502) image analyser and a Mop Video Plan semi-automated image analyser. The measurements were analysed using a one-way analysis of variance (ANOVA) where we carried out a hierarchical testing of differences between the sizes of type 1 fibers and type 2 fibers beginning with differences within and between levels in muscles, within and between muscles in rats, differences between rats (both muscles combined) within the 5 groups and differences between the 5 groups namely the 60, 90, and 120 days-old control groups, the 120 days-old hypox group, and the 120 days-old FR group.

RESULTS

Table I gives body and gastrocnemius muscle weights and shows how these weights increase with age between 60 and 120 days in normal control rats. It also shows that hypophysectomy at 60 days and food restriction begun at a similar age produce large reductions in both body and gastrocnemius muscle weights at 120 days.

Type 1 fibers (light) and type 2 fibers (dark) were analysed from each of the areas randomly selected from each of the five groups. Figures 1-3 demonstrate the treatment effects of hypophysectomy and food restriction at 120 days. Type 2 fibers were not subdivided into type 2a and 2b subgroups to avoid subjective judgments. Table II details the mean ECD values of the muscle fibers and fiber type proportions for the 5 groups of rats and demonstrates that there is very little variability in the sizes of muscle fibers within each group indicating that fiber atrophy and compensatory hypertrophy are not a feature of these relatively young animals. An increase in the ECD of fibers was recorded between the 60 and 120 days-old control groups where the type 1 ($p < 0.01$) and type 2 ($p < 0.01$) fibers grew proportionally with age. However, in the hypox group no apparent growth of the type 1 fibers occurred between 60 and 120 days ($p > 0.05$), with fiber sizes being almost identical. In contrast the type 2 fibers of the 120 days-old hypox rats were smaller in size compared with those in the 60 days-old control rats ($p < 0.01$).

In the FR group the type 1 posture fibers were retarded in growth since their mean fiber size at 120 days

TABLE I
Age, body and medial gastrocnemius muscle weights in the 5 groups of rats.

Group	Age days	Body weight g	Left and right medial gastrocnemius weight, g
Control rats	60	278.3 ± 12.07	0.63 ± 0.057
Control rats	90	380.0 ± 32.74	0.76 ± 0.091
Control rats	120	434.8 ± 22.21	0.92 ± 0.017
Hypox rats	120	173.5 ± 24.31	0.38 ± 0.062
FR rats	120	191.8 ± 11.58	0.43 ± 0.034

There were 4 rats in each group. Values are given as means ± SD.

Hypophysectomized and food restricted rats at age 120 days are about 100g lighter than controls at 60 days. This reduction in weight is mainly because of the loss of pituitary growth hormone and the involuntary halving of food intake in hypophysectomized rats. In food restricted rats food intake is likewise halved and growth hormone secretion is reduced in the long term.

approximates that recorded at 90 days of age in the control rats. The mean type 2 muscle fiber size in the 120 days-old FR rats was not significantly different ($p > 0.05$) from that recorded in the 60 days-old control rats. For each of the groups analysed, close to normal distribution of fiber sizes was found (Figures 4-6 for the 120 day old treatment groups).

The percentages of type 1 fiber types in each of the control groups, and the hypox and FR rats are detailed in Table II. There is no significant difference between the 60 days controls, 90 days controls, 120 days controls and the FR group. However, the hypox group shows a significant increase in the numbers of type 1 fibers ($p < 0.01$).

Analysis of variance of the three age groups (Table III) demonstrates significant differences ($p < 0.01$) in fiber sizes for both the type 1 and type 2 muscle fibers; these differences are within and between left and right leg muscles of each of the rats. A significant difference ($p < 0.01$) also exists in the fiber sizes for both the type 1 and type 2 muscle fibers between the leg muscles of all the rats within each of the three groups. The analysis also shows sig-

nificant effects of age for type 1 ($p < 0.05$) and type 2 ($p < 0.01$) fiber sizes.

Analysis of muscle fiber sizes between the treatment groups comprising the 120 days-old control, hypox and FR groups (Table IV) indicates that there is a significant difference ($p < 0.01$) in mean fiber size for the type 1 muscle fibers within and between the left and right leg muscles. However, between the leg muscles of all the rats within each of the groups, there is no significant difference ($p > 0.05$) in the sizes of the type 1 muscle fibers. A significant difference ($p < 0.05$) does exist in the mean fiber size of the type 1 muscle fibers between treatments. For the type 2 muscle fibers, significant differences ($p < 0.01$) in mean fiber size are apparent within and between leg muscles of the rats, between leg muscles of all the rats within each of the groups, and between treatment groups.

Further analysis of variance of the 120 days-old data (Table II) has shown that type 1 muscle fibers of the hypox group are smaller than those of the FR group ($p < 0.01$). Type 2 muscle fibers were not different in size in hypox and FR groups ($p > 0.05$). Both type 1 and 2 mus-

TABLE II
Mean equivalent circle diameter (μm) of muscle fibers and fiber type proportions
in the rat gastrocnemius muscle.

	60 days controls n=4	90 days controls n=4	120 days controls n=4	120 days Hypox. n=4	120 days FR n=4
Type 1	46.53 $\pm$ 2.84	51.06 $\pm$ 4.00 ^a	53.34 $\pm$ 4.05 ^b	46.76 $\pm$ 2.94 ^c	50.85 $\pm$ 3.17 ^d
Type 2	45.98 $\pm$ 2.59	50.75 $\pm$ 2.43 ^b	54.47 $\pm$ 3.57 ^b	41.77 $\pm$ 2.94 ^{bc}	43.31 $\pm$ 3.48 ^c
Type 1	24.5%	23.0%	24.9%	31.6% ^{bc}	21.8%

Values are given as means $\pm$ SD. Averages taken from right and left muscles from each of 4 animals from which a minimum of 18,000 muscle fibers were measured from each group and the combined averages and fiber type proportions calculated from them.

The means from each of the groups designated are statistically different:

- a from 60 days control ($p < 0.05$)
- b from 60 days control ($p < 0.01$)
- c from 120 days control ($p < 0.01$)
- d from 120 days hypox ($p < 0.01$)

cle fibers were smaller in hypox rats at 120 days than the controls of the same age ($p < 0.01$). Type 2 fibers in 120 days-old FR rats were significantly smaller than those of the 120 days-old controls ($p < 0.01$).

DISCUSSION

The results presented in this paper are a continuation of work previously published (Shorey *et al.*, 1988) on the long-term effects of hypophysectomy and food restriction in old rats aged over 1000 days. Everitt *et al.* (1980; 1985) reported that both long-term hypophysectomy and long-term food restriction prevented hind-limb paralysis in old rats. Shorey *et al.* (1988) in a morphometrical analysis on aging skeletal muscle, demonstrated considerable variability in sizes of both type 1 and type 2 fibers in old rats due to atrophy of the type 2 fibers and compensatory hypertrophy of the type 1 postural fibers. Size variability was also pronounced in the long-term food restricted rats. The progressive atrophic changes in the old control rats were

reported as probably due to a decrease in metabolic activity with increasing age resulting in limited movement and decreased usage of the type 2 fibers resulting in subsequent atrophy.

In the present study of the short-term effects of hypophysectomy and food restriction we did not observe the degree of fiber size variability (Table II) reported by Shorey *et al.* (1988) in old rats over 1000 days-old (SD type 1, 12.9; type 2, 16.1). The extent of fiber size variability in young growing animals was much less than expected at 120 days (SD type 1, 4.0; type 2, 3.6). This suggests that the increased variability of fiber size in old rats is an aging phenomenon. Short term deficiency of hormones and nutrients did not affect variability of fiber size over a 2-month period although a significant increase in the number of type 1 fibers is evident in the hypox group indicating the increased reliance on posture fibers in these sedentary rats. Only the long term actions of hormones and nutrients over more than two years were able to affect fiber size variability.

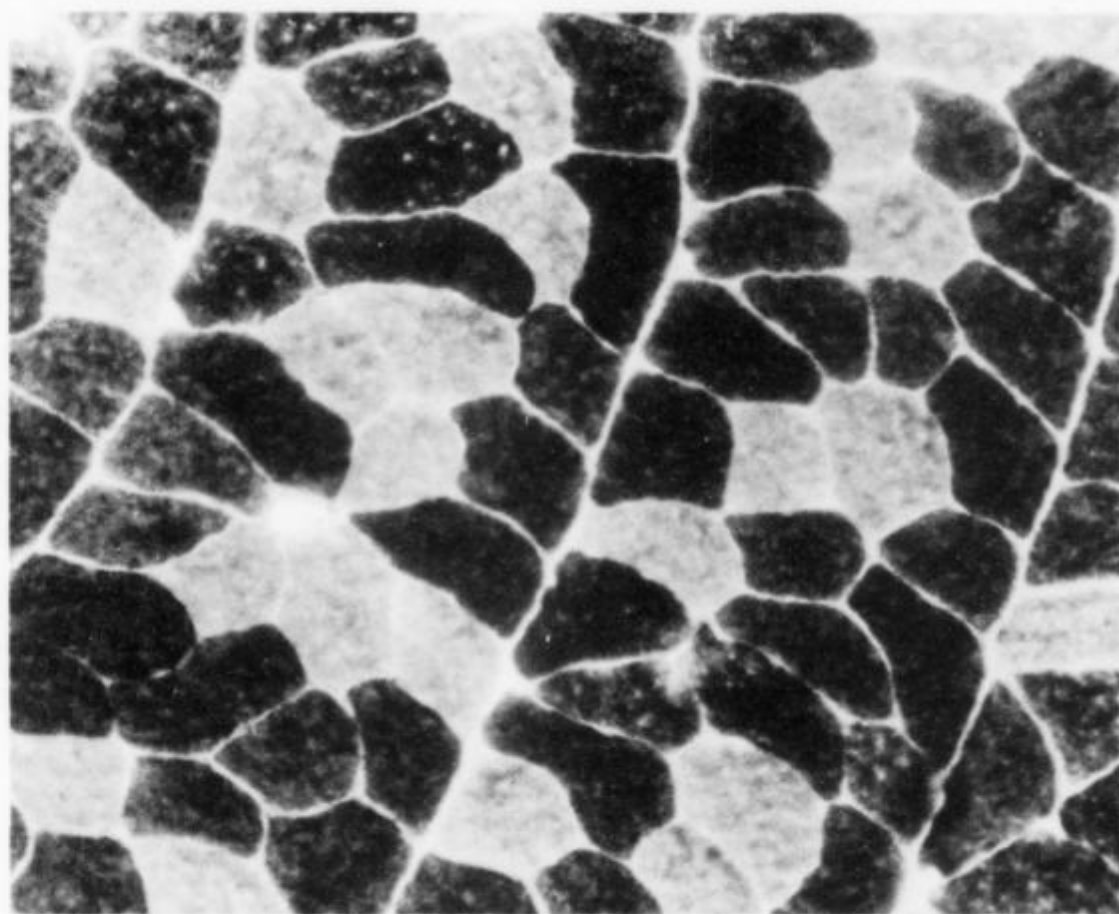


Figure 1. Transverse section of an area from the gastrocnemius muscle of a 120 days-old control rat. The magnification of Figures 1 to 3 has been standardized to X224. The type 1 muscle fibers are light in contrast, the type 2 muscle fibers are dark. These fibers are larger than those in the hypophysectomized rats (Figure 2) and the food restricted rats (Figure 3).

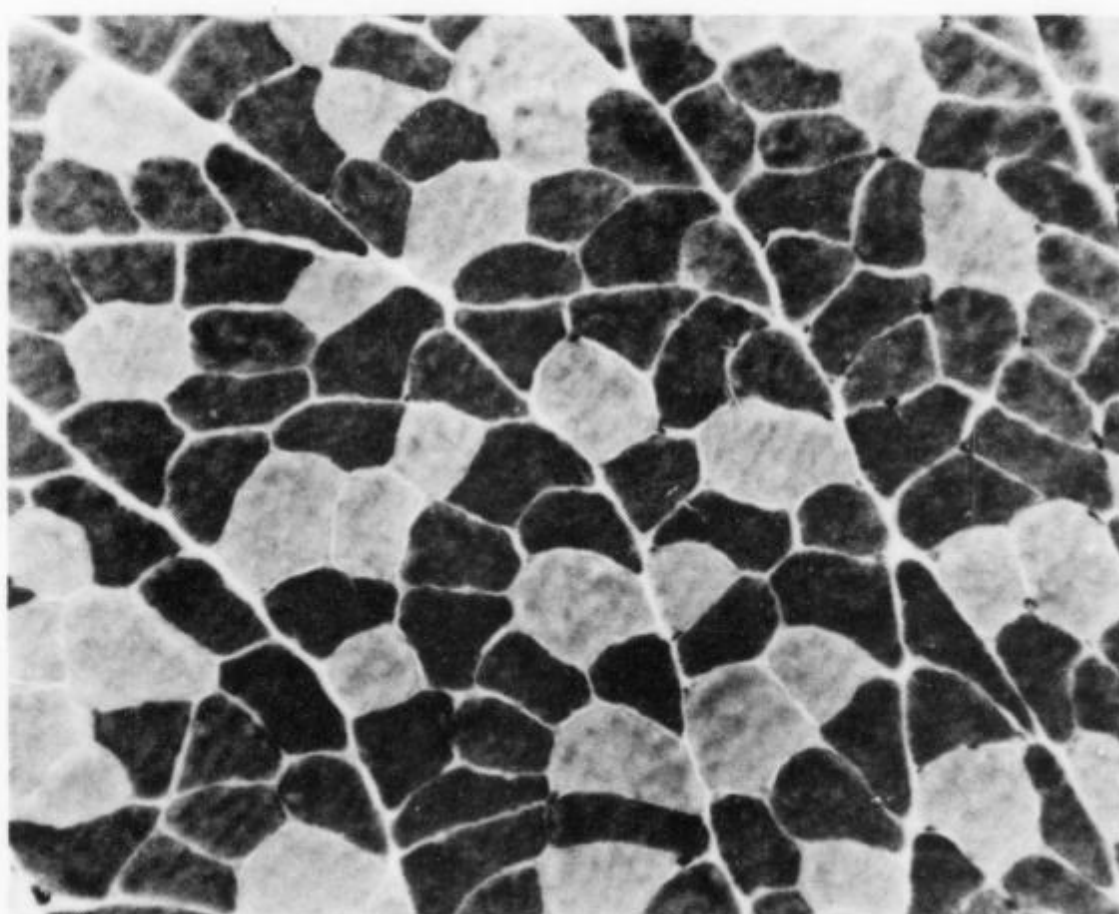


Figure 2. Transverse section of an area from a 120 days-old hypophysectomized rat. No fiber atrophy or hypertrophy is apparent.

The anterior pituitary plays an important role in influencing muscle fiber growth. Muscle fiber dependence on at least two pituitary hormones, namely growth hormone and thyroid stimulating hormone have been reported. Growth hormone increases muscle fiber size (Bigland and Jehring, 1952; Ullman and Oldfors, 1989; Ullman *et al.*, 1989) by increasing cellular proliferation and thus protein

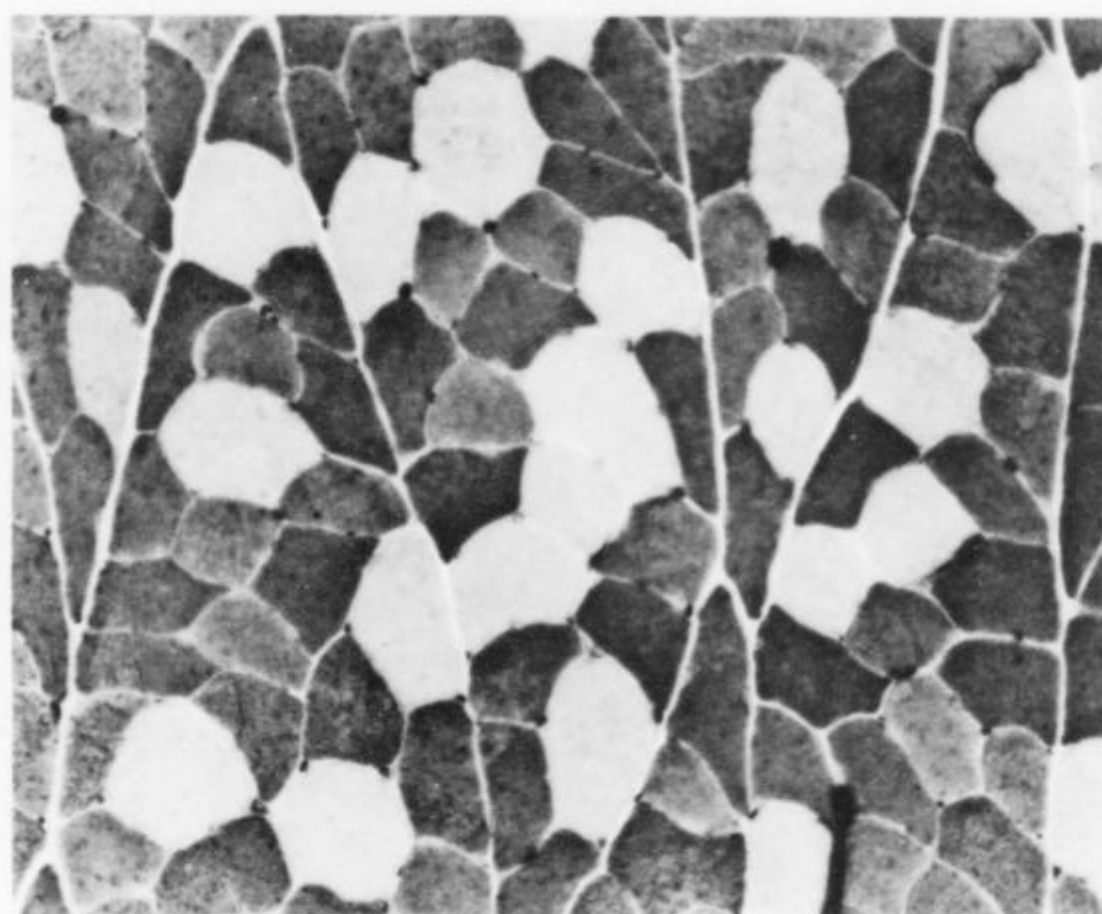


Figure 3. Transverse section of an area from a 120 days old food restricted rat. No fiber atrophy or hypertrophy is apparent.

synthesis in muscle fibers (Florini and Breuer, 1966; Ullman and Oldfors, 1989; Ullman *et al.*, 1989). Poor muscle development and function are attributed to deficiencies in the level of thyroid hormone (Florini, 1987). Thyroid hormone has been reported to affect fast fiber development (Gambke *et al.*, 1983; Salviati *et al.*, 1985). Armario *et al.* (1987) demonstrated that suppressed secretion of growth hormone and thyroid stimulating hormone occurs during food restriction. Campbell *et al.* (1977) reported increased levels of growth hormone and thyroid stimulating hormone following restoration of normal food intake in FR rats.

Low intake of nutrition also caused a decrease in the rate of protein synthesis and growth (Li and Goldberg, 1976; Millward *et al.*, 1976; Li *et al.*, 1979; Layman *et al.*, 1981). Muscle protein degradation in the food restricted rat (Li and Goldberg, 1976) also reduces growth. These investigators found that the rate of protein synthesis in muscle decreases during food restriction which could result in the retardation in growth of muscle fibers.

Normally-fed hypox rats ingest an *ad libitum* food intake similar to those on a food restricted diet of 9g /day. The differences recorded between these groups of rats is the result of some pituitary hormone action in the FR group, and its complete absence in hypox rats.

This short-term study has demonstrated that in 120 days-old hypox rats type 1 (posture) fibers develop to the size normally associated with 60 days-old control rats although there is a significant increase in the percentage of type 1 fibers (Table II). In the FR rats the type 1 fibers grow to a size normally associated with 90 days-old rats.

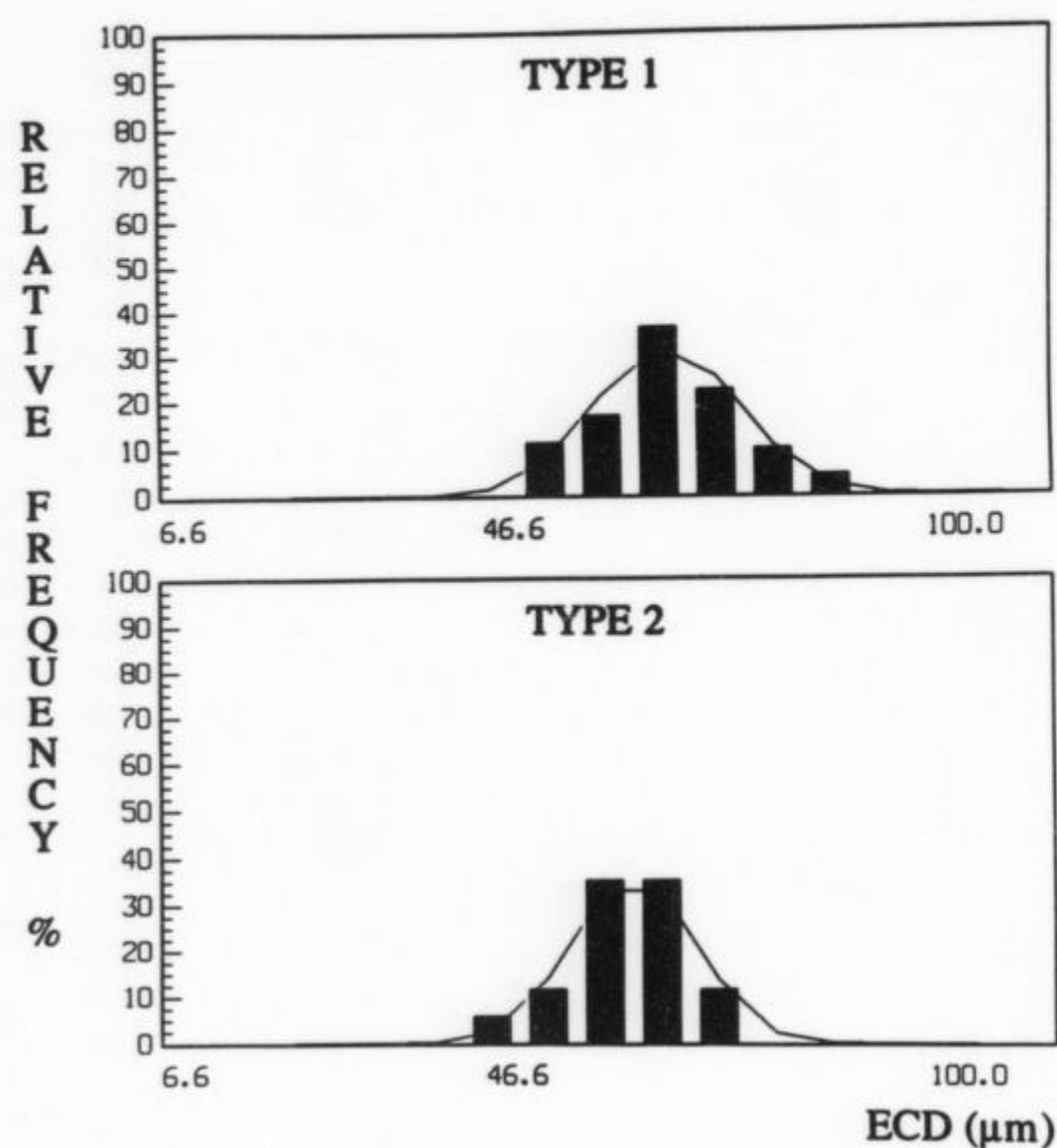


Figure 4

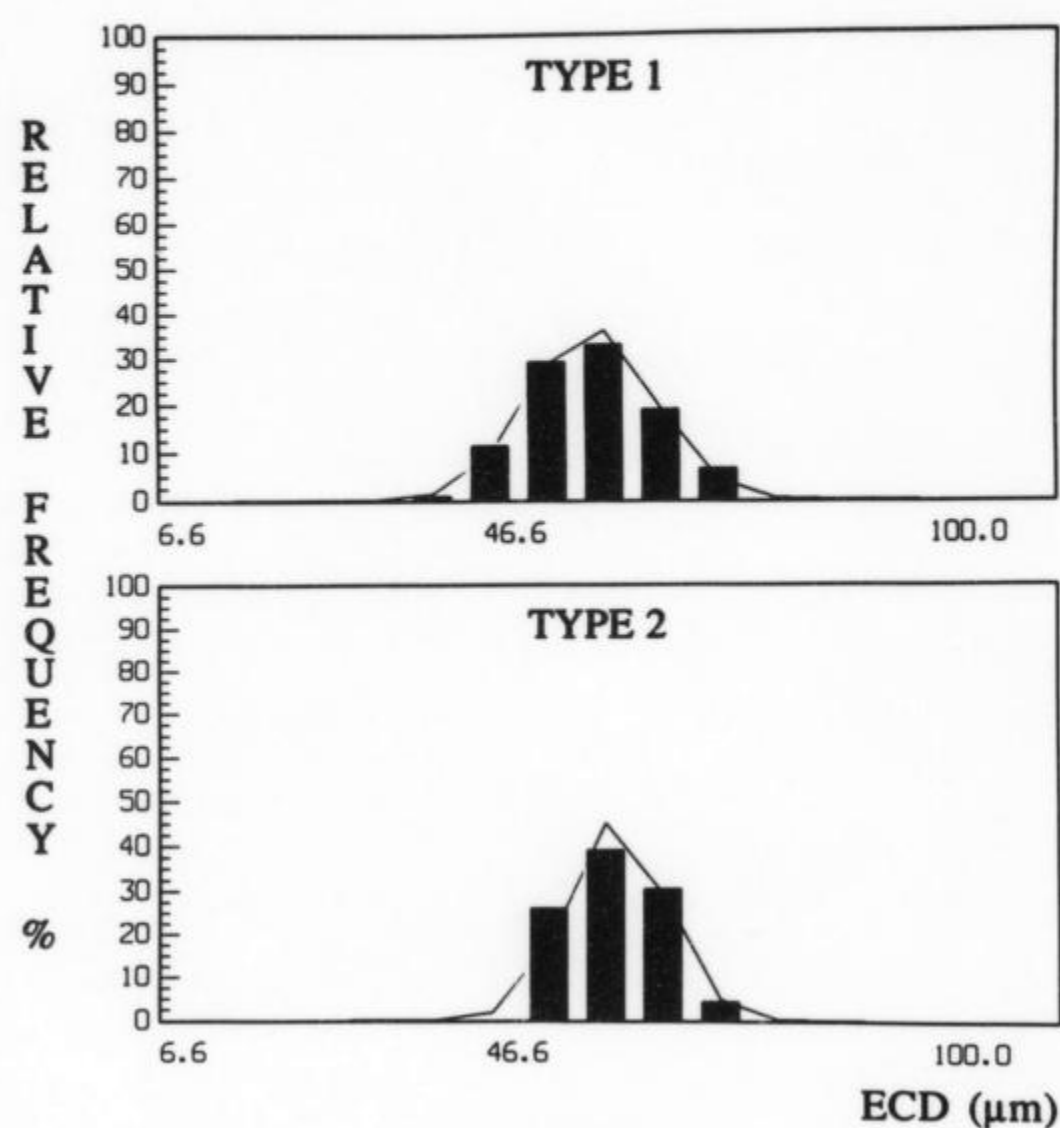


Figure 6

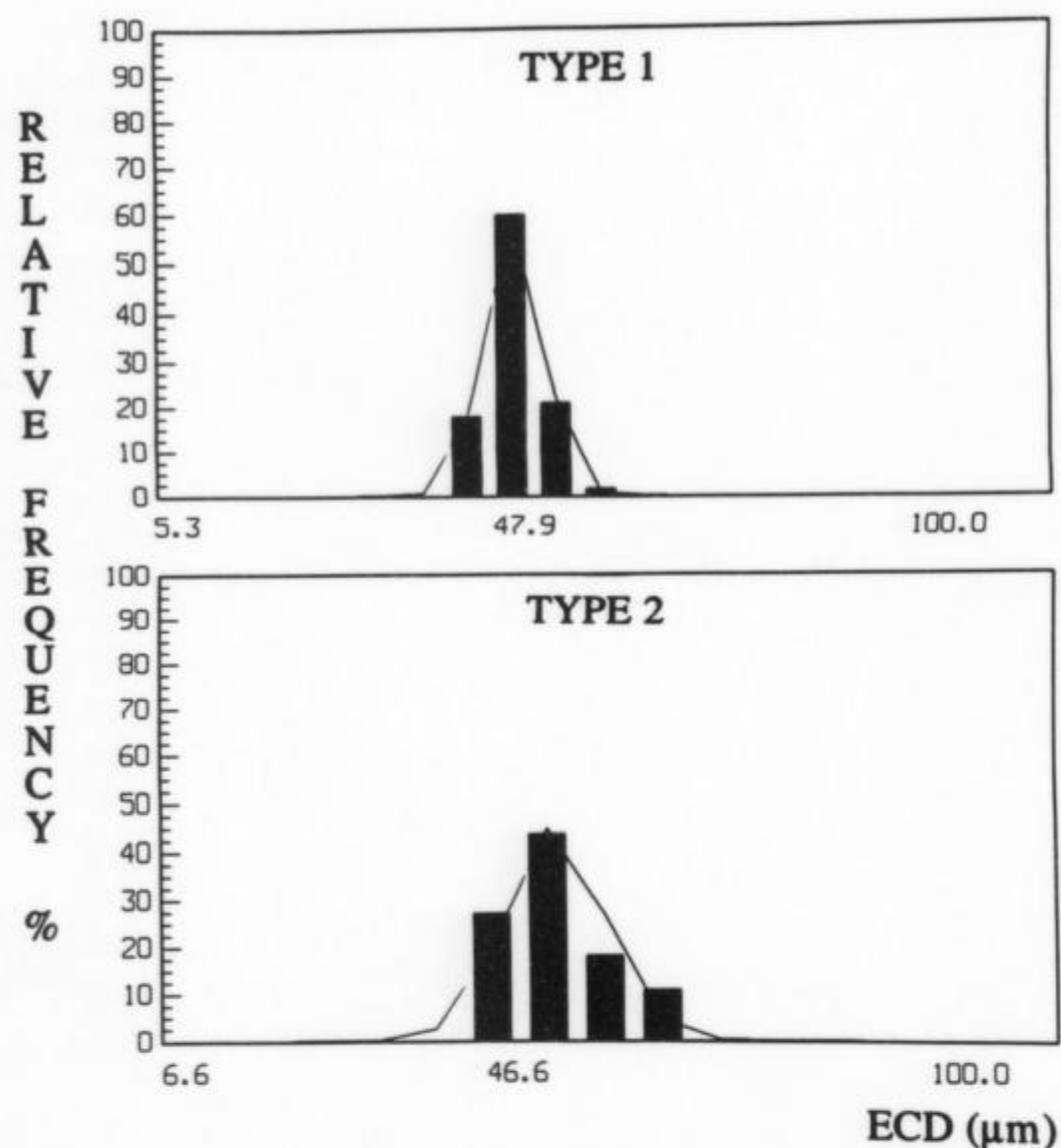


Figure 5

Figures 4-6. Distribution of the type 1 and type 2 muscle fibers of the gastrocnemius muscle of 120 days-old controls, hypophysectomized and food restricted rats respectively. Fiber cross sections were measured for equivalent circle diameter (ECD) and plotted. The range of fiber sizes in Figures 4 to 6 is calibrated from 6.6mm to 100mm with a mid-point of 46.6mm. The exception is for the type 1 fibers in Figure 5 which range in size from 5.3mm to 100mm with a mid-point of 47.9mm.

Since both hypox and FR rats were more sedentary than the controls this could partly explain this slower growth although hormonal factors also contribute.

A combination of factors appears to influence muscle fiber size and fiber type proportions; these include the level of nutrition, levels of pituitary hormones, and age together with the levels of activity which determine the

metabolic demands of the muscle fibers under varying circumstances. For instance, the type 1 (posture) fibers are in constant use since they are important for supporting the weight of the rat, irrespective of the amount of activity in which the animal is involved. In contrast, the type 2 (fast) fibers are particularly affected by old-age and food restriction because of the low level of movement observed in

TABLE III
ANOVA comparisons of age groups. Comparisons are made of type 1 and type 2 fiber sizes in 60, 90, and 120 days-old control rats.

Type 1	Degrees of Freedom	$\Sigma(x)^2$	Variance
Within muscles	192	875.8	4.6
Between muscles	12	414.2	34.5
Between rats	9	1583.8	176.0
Between ages	2	1728.5	864.2
Total	215	4602.2	
	$^{12}F_{192} = 7.6$	$p < 0.01$	
	$^9F_{12} = 5.1$	$p < 0.01$	
	$^2F_9 = 4.9$	$p < 0.05$	
Type 2	Degrees of Freedom	$\Sigma(x)^2$	Variance
Within muscles	192	768.2	4.0
Between muscle	12	235.1	19.6
Between rats	9	795.5	88.4
Between ages	2	2606.7	1303.3
Total	215	4405.4	
	$^{12}F_{192} = 4.9$	$p < 0.01$	
	$^9F_{12} = 4.5$	$p < 0.01$	
	$^2F_9 = 14.7$	$p < 0.01$	

TABLE IV

ANOVA comparisons of treatment groups at age 120 days. Comparisons are made between type 1 and type 2 fiber sizes in 120 days-old control rats, 120 days-old hypophysectomized rats and 120 days-old food restricted rats.

Type 1	Degrees of Freedom	$\Sigma(x)^2$	Variance
Within muscles	192	675.7	3.5
Between muscles	12	805.2	67.1
Between rats	9	1013.1	112.6
Between treatments	2	1591.6	795.8
Total	215	4085.6	

$$^{12}F_{192} = 19.1 \quad p < 0.01$$

$$^9F_{12} = 1.7 \quad p > 0.05$$

$$^2F_9 = 7.1 \quad p < 0.05$$

Type 2	Degrees of Freedom	$\Sigma(x)^2$	Variance
Within muscles	192	483.0	2.5
Between muscles	12	244.4	20.4
Between rats	9	1648.5	183.2
Between treatments	2	6919.2	3459.6
Total	215	9295.1	

$$^{12}F_{192} = 8.1 \quad p < 0.01$$

$$^9F_{12} = 9.0 \quad p < 0.01$$

$$^2F_9 = 18.9 \quad p < 0.01$$

these rats. Consequently these muscle fibers become atrophied through disuse. In the present study type 2 fiber atrophy was not so prominent in the FR rats (Table II) which can be explained by the low levels of pituitary hormones and increased activity, which in the FR group was sufficient to promote some growth in these fibers. At 120 days of age the range of the type 2 fiber size of the FR group was equivalent to that found in 60 days-old rats. We also observed that the mean fiber size of the type 1 muscle fibers in the gastrocnemius muscle in 120 days-old hypox rats is the same as that previously reported by Shorey *et al.* (1988) in 1000+ days-old hypophysectomized rats. This finding suggests that hypophysectomy prevents further growth of type 1 muscle fibers during the 2 months period of this study although there is a significant increase in the number of type 1 fibers together with a progressive and parallel increase in both muscle and body weights.

In conclusion it is apparent that any effects on muscle fiber size development resulting from hypophysectomy or food restriction are initiated early during postnatal development of the gastrocnemius muscle. Lack of normal levels of pituitary hormones may be a contributing factor to this lack of muscle fiber growth. The early effects of pituitary hormones are concerned with muscle fiber growth, while the life-long effects relate to aging changes such as muscle fiber atrophy.

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REFERENCES

- ARMARIO, I., MONTERO, J.L., JOLIN, T. 1987. Chronic food restriction and the circadian rhythms of pituitary-adrenal hormones, growth hormone and thyroid stimulating hormone. *Ann. Nutr. Metab.*, 31, 81-87.
- BIGLAND, B. & JEHRING, B. 1952. Muscle performance in rats, normal and treated with growth hormone. *J. Physiol.*, 116, 129-136.
- CAMPBELL, G.A., KURCZ, M., MARSHALL, S., MEITES, J. 1977. Effects of starvation in rats on serum levels of follicle stimulating hormone, luteinizing hormone, thyrotropin, growth hormone and prolactin; Response to LH-releasing hormone and thyrotropin-releasing hormone. *Endocrinol.*, 100, 580-588.
- EVERITT, A.V., SEEDMAN, N.J., JONES, F. 1980. The effects of hypophysectomy and continuous food restriction, begun at ages 70 and 400 days, on collagen aging, proteinuria, incidence of pathology and longevity in the male rat. *Mech. Age. Dev.*, 12, 161-172.
- EVERITT, A.V., SHOREY, C.D., FICARRA, M.A. 1985. Skeletal muscle aging in the hindlimb of the old male Wistar rat: Inhibitory effect of hypophysectomy and food restriction. *Arch. Gerontol. Geriatr.*, 4, 101-115.
- EVERITT, A.V. & WYNDHAM, J.R. 1982. Hypothalamic-pituitary regulation and aging. In *Endocrine and Neuroendocrine Mechanisms of Aging*, (eds. R.C. Adelman & G.E. Roth), pp 123-165. CRC Press, Boca Raton FL.
- FLORINI, J.R. 1987. Hormonal control of muscle growth. *Muscle and Nerve*, 10, 577-598.
- FLORINI, J.R. & BREUER, C.B. 1966. Amino acid incorporation into protein by cell-free systems from rat skeletal muscle. V. Effects of pituitary growth hormone on activity of ribosomes and ribonucleic acid polymerase in hypophysectomized rats. *Biochem.*, 5, 1870-1876.
- GAMBKE, B., LYONS, G.E., HASELGROVE, J., KELLY, A.M., RUBINSTEIN, N.A. 1983. Thyroidal and neural control of myosin transitions during development of rat fast and slow muscles. *FEBS Lett.*, 156, 335-339.
- GUTH, L. & SAMAHA, F.J. 1969. Qualitative differences between actomyosin ATPase of slow and fast mammalian muscles. *Exp. Neurol.*, 25, 138-152.
- KOYAMA, R. 1962. A simple method of hypophysectomy in rats. *Endocrinol. Jpn.*, 9, 321-323.
- LAYMAN, D.K., SWAN, P.B., HEGARTY, P.V.J. 1981. The effects of acute dietary restriction on muscle fibre number in weaning rats. *Br. J. Nutr.*, 45, 475-481.
- LI, J.B. & GOLDBERG, A.L. 1976. Effects of food deprivation on protein synthesis and degradation in rat skeletal muscles. *Am. J. Physiol.*, 231, 441-448.
- LI, J.B., HIGGINS, J.E., JEFFERSON, L.S. 1979. Changes in protein turnover in skeletal muscle in response to fasting. *Am. J. Physiol.*, 236, E222-E228.
- MILLARD, D.J., GARLICK, P.J., NNANYELUGO, D.O., WATERLOO, J.C. 1976. The relative importance of muscle protein synthesis and breakdown in the regulation of muscle mass. *Biochem. J.*, 156, 185-188.
- PADYKULA, H. & HERMAN, E. 1955. The specificity of the method for adenosine triphosphatase. *J. Histochem. Cytochem.*, 3, 170-183.
- SALVIATI, G., ZEVIANI, M., BETTO, R., NACAMULLI, D., BUSNARDO, B. 1985. Effects of thyroid hormones on the biochemical specialization of human muscle fibers. *Muscle and Nerve*, 8, 363-371.
- SHOREY, C.D., MANNING, L.A., EVERITT, A.V. 1988. Morphometrical analysis of skeletal muscle fibre ageing and the effect of hypophysectomy and food restriction in the rat. *Gerontol.*, 34, 97-109.
- ULLMAN, M. & OLDFORS, A. 1989. Effects of growth hormone on skeletal muscle. 1. Studies on normal rats. *Acta Physiol. Scand.*, 135, 531-536.
- ULLMAN, M., ALAMEDDINE, H., SKOTTNER, A., OLDFORS, A. 1989. Effects of growth hormone on skeletal muscle. 11. Studies on regeneration and denervation in adult rats. *Acta Physiol. Scand.*, 135, 537-543.